

Who is the Better Operator of an Identity Wallet Prioritised by the User?

A Quantitative Survey Between State and Company

Sandra Kostic

sandra.kostic@aisec.fraunhofer.de

Fraunhofer Institute for Applied and Integrated Security AISEC
Garching near Munich, Germany

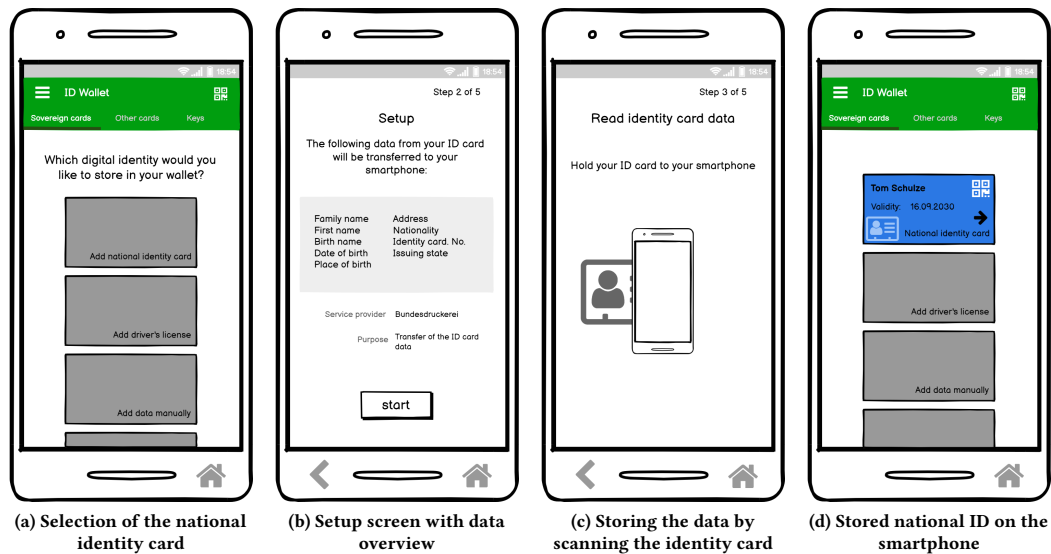


Figure 1: Creation of the digital national ID card and storage in the Identity Wallet app
(Note: These images do not represent the entire process.)

ABSTRACT

This paper presents the results of an online survey based on a developed concept that enables the storage of multiple digital identities in one app. Users manage these identities in the so-called Identity Wallet independently and decide which data should be sent to any external service. As previous usability studies indicate that the operator of the Identity Wallet is decisive for user trust in such technologies, a quantitative study was conducted with 306 people. The participants were asked to rank possible operators (e.g. state, open source community projects, known company, etc.) in order of their preference. The results show that, on average, a private company is favoured as the operator. However, if the operator that was named first by most participants is considered, then this is the state.

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CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

KEYWORDS

Digital Wallet, Digital Identity, Survey, User Trust, Operator

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1 INTRODUCTION

With the help of the first name, surname and date of birth, it is possible to uniquely identify a person [16]. In order to be able to identify people online, so-called digital identities are required. Digital identities can already be found in services such as Facebook or Google (Login with Facebook [10] or Google [4]) [11]. Users of these services can create an account with their personal data and

use this same account – their digital identity issued by Facebook or Google – to identify themselves to other services.

An alternative to this is an Identity Wallet [6, 13, 14]. It makes it possible to store many different digital identities in a single app. The user has sole control over their data and decides for themselves which data they want to share with which service provider, e.g. to identify themselves or to share proof of age without further information [13].

In order to meet the requirements of future users of an Identity Wallet, especially since it turned out that already available Identity Wallet solutions had problems with usability at the time [5, 14], a concept for an Identity Wallet was developed as a low-fidelity prototype based on a focus group in 2020 [7] (e.g. to demonstrate how a national ID card or driving licence can be stored in the Identity Wallet) (see Figure 1¹). This prototype was not only used to visually illustrate an application. It was also to be used for user studies to investigate both the usability and comprehensibility of the concept [7].

Over a period of two years (between 2020 and 2022), three qualitative user studies were conducted with a total of 44 people [8]. Besides the finding that the participants were able to create a digital identity with the prototype and showed great interest in using the Identity Wallet, the results also indicated that the *operator of the Identity Wallet* is crucial for users' trust in the concept [8]. As the wallet is used to manage personal data, it was very important to take the trust aspect into account and address any concerns.

The first user study was conducted in 2020 with the low-fidelity prototype. Just over half of the participants thought that the government was the right operator for the wallet. The rest favoured a company as the operator. The prototype was revised into a high-fidelity prototype in 2022 using the feedback collected and the study was repeated. A large majority were in favour of the government as the operator, with only one person favouring a company. To rule out the possibility that the change in users' preference for the wallet operator was influenced by design or visual impression the high-fidelity prototype, the 2020 study was replicated with the low-fidelity prototype at the end of 2022. The majority of respondents again favoured the state as the operator, while a small number preferred a company [8]. This indicates a temporal shift in users' preference for the wallet provider.

To continue the research, two research questions had to be investigated:

- RQ1: What preferences do users show regarding the operator of an Identity Wallet when they cannot interact with the Identity Wallet prototype?
- RQ2: What influence does the use case of the digitalisation of the national ID card have on the preference of an operator?

In order to answer these questions, a quantitative study was conducted in Germany with 306 people, in which the participants were asked to indicate a preference among given operators (e.g. state, known private company, open source project, etc.). Furthermore, the participants were only shown images of the low-fidelity prototype.

¹The screens have been translated into English for this publication. The original prototype is in German.

In addition, the selected images of the original prototype were edited so that no reference to a sovereign ID (e.g. national ID card) could be seen.

As a result, when asked about the preferred operator, a known private company was named as the preferred operator on average across all preferences mentioned. However, looking at the responses, the operator most frequently named as preference 1 was the state. This indicates that although no sovereign ID (national ID card) was presented, there is still a strong preference for the state.

Briefly summarised, this publication can make the following contribution:

- A Quantitative survey on a prototype developed based on user requirements
- Insight into the preferences of potential Identity Wallet operators from a user perspective with a focus on user trust
- Insights regarding the Identity Wallet's support of sovereign and non-sovereign documents and influence on a user's preference for an operator

2 RELATED WORK

As current applications for Identity Wallets are still at an early stage, there are some challenges in their use [2, 5, 6, 14]. Research has shown that the innovation of Identity Wallets and the benefits of using them have not been sufficiently explained to users [5, 14]. This is said to have led to significant problems and impaired the use of the technology.

In particular, the usability of Identity Wallet was analysed. For instance, Khayretdinova et al. [5] conducted a usability study with 18 people who used various Identity Wallets based on SSI technology [2]. The authors found that at the time of the study, the technology did not offer users the opportunity to experience the benefits of an Identity Wallet in terms of privacy or security [5]. Furthermore, there are indications that the usability of Identity Wallets is said to have suffered due to the poor description of the terminology used [5, 14]. Usability problems are also said to have led to users expressing concerns about sharing too much data [6].

"You know I work in information technology already and part of me says the idea that you keep your information secure and people not knowing it is a ship that has probably already sailed" [6]

According to Khayretdinova et al [5], as Identity Wallet is a new concept and technology, it is important to provide clear explanations of the functions of the Identity Wallet, including how to use these functions.

Sjöholm et al. [15] also addressed the issue of trust in Identity Wallets apart from a classic user study. Again, their findings indicate how crucial the usability of the Identity Wallet is for users. Users are said to have considered usability more important than security and privacy. In order to strengthen trust, security nets should also be used in the Identity Wallet. In addition, people in different Member States of the European Union are said to have different cultural expectations in terms of trust, which should be taken into account when it comes to providing such solutions.

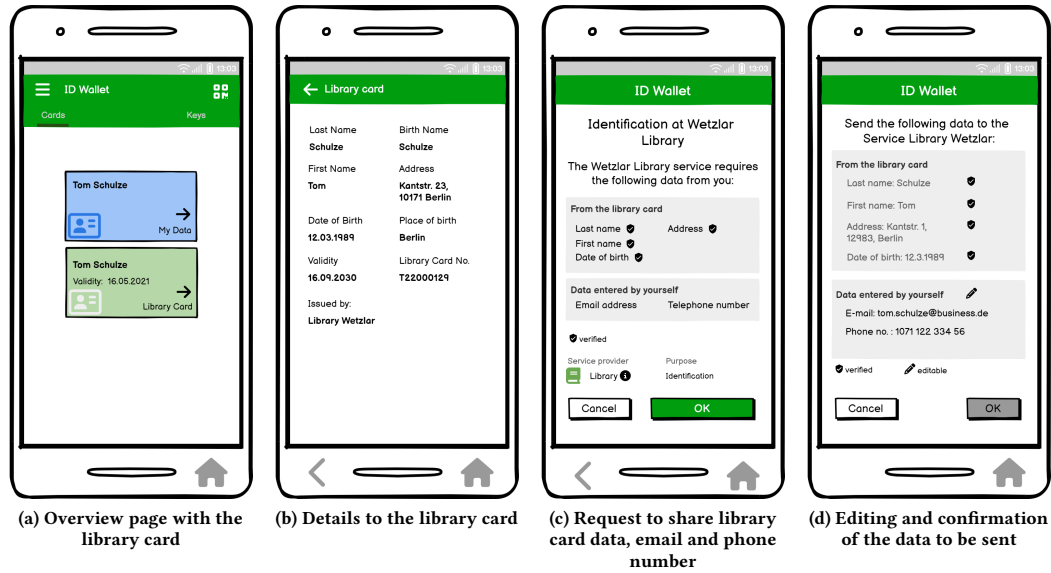


Figure 2: Process flow for requesting and sending library card data

3 IDENTITY WALLET CONCEPT

An Identity Wallet enables the user to handle their own digital identities independently [6], so that the user alone decides which data the requesting service will receive, e.g. for the purpose of identification.

The developed Identity Wallet concept [7] covers the following core functions:

- Independent storage of digital identities by transferring the data on the physical ID card to the smartphone app (e.g. national ID card see Figure 1) or by transferring a digital proof and storing it in the wallet provided by a service (e.g. digital library card see Figure 2a)
- Use of the digitally stored identities (e.g. for identification see Figure 2c und 2d) by transmitting the data to the requesting service. The user alone decides whether the data is transferred and which exact data is transferred. Data transmission is only authorised with the user's consent.

4 QUANTITATIVE SURVEY

4.1 Structure of the questionnaire

The idea of the survey was that participants in the study would familiarise themselves with the developed concept of the Identity Wallet app. In particular, it was pointed out that the participants in the study did not need any prior knowledge to answer the survey and were also instructed not to research unknown information during the survey. The aim was to find out to what extent the concept is self-explanatory.

For this purpose, four screens were selected from the interactive prototype with descriptive text² with the aim of reflecting the

essence of the Identity Wallet (see Section 3), namely the management and sharing of data by the user. Further information on the developed concept of the Identity Wallet was not provided.

The selected screens included the following:

- (1) Screen 1: Overview page with the digital identities already stored (see Figure 2a).

The following text was given in the survey to explain the given image:

"You can think of the wallet as a kind of secure, virtual wallet. You can store virtual ID documents in it. In this example, you have already been issued a virtual library card."

- (2) Screen 2: Detailed view of the saved library card (see Figure 2b)

The following text was given in the survey to explain the given image:

"ID documents can be issued to you by issuers. They contain information that has been verified by the respective issuer. In addition to traditional ID documents (such as the library card in the example), digital keys and tickets can also be stored in the wallet. In the example given, you have been issued a library card by the Wetzlar Library. This contains verified information about you as the card holder."

- (3) Screen 3 and 4: Data request from Wetzlar Library and confirmation that the data is to be sent (see Figure 2c and 2d). The following text was given in the survey to explain the given image:

²Note: The descriptive texts are originally in German and have been translated into English for this paper.

Table 1: Age distribution of the participants

< 30 Years	30-39 Years	40-49 Years	50-59 Years	>=60 Years	Average age	Total
72	72	61	58	43	42	306

"When you identify yourself with your ID documents, you will receive a request to share data (Figure 2c). This data can be obtained from your ID documents. You can select which data is to be transmitted (Figure 2d). Depending on the application, it is also possible to transfer only parts of the requested data. It may also be possible to enter missing data manually if it does not require verification.

In the example shown, your first and last name, your address and your date of birth were requested. These were taken from your library card. You could also enter your e-mail address and telephone number."

It should be noted that, unlike the original prototype [7], these screens in Figure 2 were customised specifically for this online survey. All references to a digitalised national ID card were removed and replaced with a request for a library card.

The reason for this was to prevent a possible bias in favour of the state, as users see that national ID cards can also be used with the Identity Wallet. In previous studies, it had been shown that, due to the fact that the national ID card is provided by the state, the participants stated that this must also apply to the digitised version of the national ID card [7].

After the participants were able to familiarise themselves with the screens at their own pace, they were asked which operator they would prefer for the application presented. The participants were asked to prioritise the named operators (priority 1 to 7), starting with the highest priority (priority 1) and ending with the lowest priority (priority 7).

The given operators to be prioritised were as follows:

- Known German Private Company
- Known non-German Private Company
- Unknown German Private Company
- Unknown non-German Private Company
- Community Project Open Source
- German Government
- Known Organisation (e.g. the Chaos Computer Club (CCC)³, etc.)
- Other (free text field)

In comparison to previous studies, in which only a preference for a state or private operator was asked [7, 8], this survey was intended to offer respondents a broader choice. In addition, the aim was to determine the extent to which the level of awareness or the location of the company (in Germany, outside Germany) had an influence on the results.

Finally, in addition to demographic questions, the participants in the study were also asked to what extent their prioritisation

would change if they were told that the Identity Wallet could also be used with sovereign documents such as a digital national ID card and a digital driving licence. This question was again intended to determine the influence of the supported sovereign identity documents on the choice of operator.

4.2 Participants and recruitment

Given that the research work was funded by the German Ministry of Economics and digital identities are currently a highly debated topic in Germany[3, 17], the study was conducted exclusively with German people.

The participants in the study were acquired via the TestingTime⁴ acquisition platform. A total of 300 people aged between 18 and 75 were approached in order to obtain a broad range of feedback. Care was taken to ensure that the participants were evenly represented in terms of age and gender.

The participants were not required to have any prior knowledge, as the aim was to find out how self-explanatory the product itself is. Furthermore, no experts in the fields of security or usability were asked or people who are specifically familiar with digital identities and Identity Wallets.

The participants received an incentive payment of €20. On average, participants took around 10-12 minutes to complete the survey.

Table 2: Order of preferred operators according to average rating

Ranking	Operator	Average rating
1	Known German Private Company	2,53
2	German Government	3,2
3	Known Organisation	3,3
4	Known non-German Private Company	3,9
5	Community Project Open Source	4,6
6	Unknown German Private Company	4,7
7	Unknown non-German Private Company	5,8

4.3 Results

Although 300 people were to be recruited via the Testing Time acquisition platform, responses were received from 306 different people (see Table 1).

The average age of the respondents was 42. The youngest person was 19 years old. The oldest was 72 years old. Overall, just as many

³The Chaos Computer Club e. V. (CCC) is the largest European hacker organisation and publicises important problems in applications' IT security. <https://www.ccc.de/en/> Accessed: January 17, 2024

⁴<https://www.testingtime.com/en/> Accessed: January 17, 2024

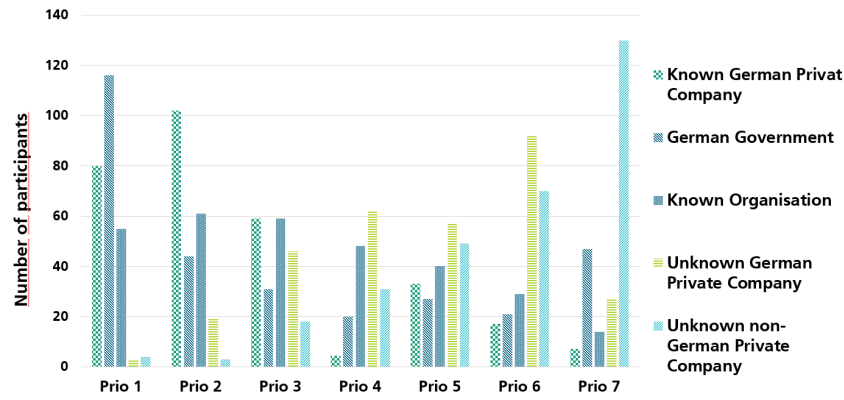


Figure 3: Diagram illustrating the priorities of the operators in relation to each other

people who identified themselves as women took part in the study as people who identified themselves as men.

When asked to prioritise the named operators of an Identity Wallet, the following order (see Table 2) was determined based on the average rating (starting with the highest priority and ending with the lowest priority). These results were calculated using the findings from Table 3.

If focus is placed on the operator that most respondents categorised as priority 1, this is by far the German government with 116 responses (see Figure 3 and Table 3).

A well-known German private company follows in second place with 80 responses. The known Organisation is once again in third place.

The lowest rating was given to the unknown non-German private company as operator. It came in last place with 130 (see Table 3), both in the average rating (see Table 2) and in the specific prioritisation (see Figure 3). This is the highest score given across all priorities.

When asked to what extent the operators' prioritisation would change after it was pointed out that sovereign documents such as the national ID card are also supported by the Identity Wallet, only 12 out of 306 people stated that they would change their prioritisation. A clear majority did not wish to make any changes.

4.4 Discussion

Based on the two research questions (see Section 1), the following conclusions can be drawn from the results.

With regard to the first research question (see RQ1 in Section 1)), which preference a user shows for an identity wallet operator, based on screens of a developed interactive prototype, it can be assumed that the screens with the provided descriptions alone were sufficient to create a basis for expressing a preference for an operator.

This assumption is supported by comments made by the participants in the free text fields.

For example, concerns were expressed about the protection of stored data as well as sent data or about forgetting a wallet password. One participant said:

Table 3: Overview of results for the prioritised operators of an Identity Wallet

Known German Private Company						
Prio 1	Prio2	Prio 3	Prio 4	Prio 5	Prio 6	Prio 7
80	102	59	33	17	8	7
Known non-German Private Company						
Prio 1	Prio2	Prio 3	Prio 4	Prio 5	Prio 6	Prio 7
21	46	62	66	54	43	14
Unknown German Private Company						
Prio 1	Prio2	Prio 3	Prio 4	Prio 5	Prio 6	Prio 7
3	19	46	62	57	92	27
Unknown non-German Private Company						
Prio 1	Prio2	Prio 3	Prio 4	Prio 5	Prio 6	Prio 7
4	3	18	31	49	70	130
Community Project Open Source						
Prio 1	Prio2	Prio 3	Prio 4	Prio 5	Prio 6	Prio 7
27	31	31	46	62	42	66
German Government						
Prio 1	Prio2	Prio 3	Prio 4	Prio 5	Prio 6	Prio 7
116	44	31	20	27	21	47
Known Organisation						
Prio 1	Prio2	Prio 3	Prio 4	Prio 5	Prio 6	Prio 7
55	61	59	48	40	29	14

"I have little interest in the wallet because you often use your smartphone quickly and information is often sent out that should not be sent out. In this case, I would really only use the wallet as a card holder without sharing"⁵

Or a comment such as:

"Such digital services harbour a higher risk of being affected by data misuse compared to similar services in the public sector"⁶

Statements like these suggest that the concept of the Identity Wallet has been correctly understood and that a good assessment of a preferred operator could therefore be made.

Both the known private company and the state are the two most important operators of an Identity Wallet for the respondents. Even though the known private company was rated as the highest priority on average, the state was given priority 1 by most participants in the study. This indicates that these two together could also be a suitable operator from the user's point of view and could therefore satisfy most people. Similar conclusions could also be drawn from earlier studies.

According to Kostic et. al. [8], the participants in this study stated that in this constellation, the company could take over the implementation work, while the state provides financial support and ensures compliance with certain security and data protection standards and regulations.

Regarding the second research question (see RQ2 in Section 1), although the Identity Wallet screens did not indicate the use of a sovereign ID card (such as the national ID card), the participants gave a high priority to a state operator.

This suggests that regardless of which (sovereign or non-sovereign) identity is stored in a wallet, the participants in the studies see a strong anchor of trust in a state operator or consider it to be very competent. It can also be assumed that the support of sovereign or non-sovereign documents in the Identity Wallet has no influence on the prioritisation of a particular operator. The participants in the study appear to make a decision in favour of a suitable operator regardless of this.

Furthermore, the results of the study indicate that awareness of a company is of great importance to the respondents. It seemed to be more relevant that the company is well-known than that it is based in Germany (see prioritisation list in Section 4.3). However, it is unclear which companies the respondents were thinking of when they rated the known companies higher. It is also unclear why a well-known company is considered to have better expertise as an Identity Wallet operator.

5 LIMITATION AND FUTURE WORKS

As digital identities and Identity Wallets are currently a relevant topic in Germany [3], the studies were conducted with German participants. Moreover, this research was funded by the German Ministry of Economic Affairs.

⁵Note: The original comment was in German and has been translated into English for this publication.

⁶Note: The original comment was in German and has been translated into English for this publication.

However, it is conceivable to extend the study to outside Germany in order to evaluate different mental models and preferences for the state or a private company. As there are endeavours at European level in particular with regard to an EU Wallet [1], this is planned as the next step.

As there was still a West Germany and an East Germany in Germany a little more than 30 years ago [9], and thus different attitudes and mental models of citizens towards the state [12], it is also conceivable to include further demographic questions in future studies. For example, it would be conceivable to determine the place of birth or the place where the longest period of life was spent, and thus the extent to which a tendency towards or against the state can be determined.

So far, this study has only identified that known companies are highly valued by the participants. However, it remains to be determined which specific companies were categorised as known by the participants. In addition, it should also be determined why the reputation of a company is more decisive for the user than the location of the company's headquarters.

The existing studies are based on prototypes. An extension of the study to analyse the interaction with a real demonstrator would therefore also be considered. This should be done as soon as corresponding applications with a real use case exist.

6 CONCLUSION

This paper presents the results of a quantitative study on a concept developed for an Identity Wallet that enables users to store and manage various digital identities independently in one app. The study captures the preferred order (in descending order of preference) of an Identity Wallet operator from the user's perspective, as it was determined that the Identity wallet operator has a strong influence on user trust in this technology.

The results show that, on average, a well-known private company is favoured across all responses. However, when looking at the ranking, most respondents put the state in first place as the operator.

It can also be assumed that the preference for a state operator also applies regardless of the possibility of storing sovereign identities, such as the national ID card.

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